Lecture 7. Optimization

Golden section, Newton, line search, Fisher scoring, GLM and IRLS, Gauss–Newton, gradient and coordinate descent

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We consider $\min_{x\in\mathbb{R}^d} f(x)$; maximization is minimization of $-f$.

7.1 Golden section search

For $d = 1$, let f be continuous and **strictly unimodal** on $[a, b]$: $f(\lambda x+(1-\lambda)y) < \max\{f(x), f(y)\}$ for $x \neq y$, $\lambda \in (0, 1)$. Keep three points $a < x_1 < b$ with $f(x_1) < \min\{f(a), f(b)\}$, so the minimum lies in (a, b). Place a new point x_0 in the longer of (a, x_1) and (x_1, b); say $a < x_0 < x_1$.

- If $f(x_0) > f(x_1)$, the minimum is in (x_0, b): set $(a, x_1, b) \leftarrow (x_0, x_1, b)$.
- Otherwise it is in (a, x_1): set $(a, x_1, b) \leftarrow (a, x_0, x_1)$.

To make both outcomes equally good and keep the configuration self-similar, require $b-x_0 = x_1-a$ and that the interior point divides the new interval in the same proportion. With $\alpha = (x_0-a)/(b-a)$ and $y = (x_1 - x_0)/(b - a)$, this gives $\alpha + y + \alpha = 1$ and $1 : \alpha = (y + \alpha) : y$, whose solution is

$$\alpha = \frac{3-\sqrt{5}}{2}, \qquad 1-\alpha = \frac{\sqrt{5}-1}{2} \approx 0.618,$$

the golden section. Each step shrinks the interval by the factor 0.618 with one function evaluation: slow (linear) but guaranteed to find the global minimum of a unimodal function.

7.2 Newton's method

At a minimizer, $\nabla f(x^\star) = 0$. Applying Newton–Raphson to ∇f is equivalent to minimizing the local quadratic model

$$f(x) \approx f(x^{(t)}) + \nabla f(x^{(t)})^\top (x - x^{(t)}) + \tfrac{1}{2}(x - x^{(t)})^\top \nabla^2 f(x^{(t)})(x - x^{(t)}),$$

giving the **naive Newton** iteration

$$x^{(t+1)} = x^{(t)} - [\nabla^2 f(x^{(t)})]^{-1} \nabla f(x^{(t)}).$$

It converges quadratically near a minimizer with positive definite Hessian. But it is not a descent method: it is equally happy to go uphill and can converge to a maximum or a saddle, or diverge, depending on the start (try $f = \sin$).

7.3 Practical Newton: positive definite approximations and line search

Replace the Hessian by a positive definite $H^{(t)}$ and choose a step size s:

$$x^{(t+1)} = x^{(t)} + s\,\Delta x^{(t)}, \qquad \Delta x^{(t)} = -[H^{(t)}]^{-1} \nabla f(x^{(t)}).$$

Proposition 7.1 (Descent direction)

If $H^{(t)}$ is positive definite and $\nabla f(x^{(t)}) \neq 0$, then $f(x^{(t)} + s\Delta x^{(t)}) < f(x^{(t)})$ for all sufficiently small $s > 0$.

Proof. By Taylor's theorem, $f(x^{(t)}+s\Delta x^{(t)})-f(x^{(t)}) = -s\,\nabla f^\top [H^{(t)}]^{-1}\nabla f + o(s)$, and the quadratic form is positive. ∎

($\{\nabla f^\top H^{-1} \nabla f\}^{1/2}$ is the **Newton decrement**.) For strictly convex f the true Hessian already works (**damped Newton**), but a line search is still needed for global convergence. Taking $H^{(t)} = I$ gives gradient descent.

Line search. Given the direction, choose s: exactly (minimize $f(x+s\Delta x)$, e.g. by golden section), or approximately by step halving or **backtracking** (Armijo): with $\alpha \in (0, \frac{1}{2})$ and $\beta \in (0,1)$, start from $s = 1$ and set $s \leftarrow \beta s$ while

$$f(x + s\Delta x) > f(x) + \alpha s \nabla f(x)^\top \Delta x.$$

The condition accepts s once f lies below a line whose slope is a fraction α of the initial slope, which always happens for small s by Proposition 7.1. The direction is computed once per iteration; the line search costs only function evaluations. A common implementation slip: if the trial point $x_s = x + s\Delta x$ is formed first, the right side of the test is $f(x) + \alpha \nabla f(x)^\top (x_s - x)$, which already contains s; writing $\alpha s \nabla f(x)^\top (x_s - x)$ counts s twice and makes the test too lenient. (For IRLS, the trial point for step s can be computed by refitting the weighted least squares with working response $z = \eta + sW^{-1}(y - \mu)$, since the fit is linear in z and returns exactly $\beta + s\Delta\beta$.)

7.4 Fisher scoring

For maximum likelihood, $f = -\ell(\theta)$. **Fisher scoring** replaces $-\nabla^2 \ell$ by the expected information

$$I(\theta) = E[-\nabla^2 \ell(\theta)] = E[\nabla \ell(\theta) \nabla \ell(\theta)^\top] \succeq 0,$$

(the equality holds when differentiation and expectation can be interchanged), giving

$$\theta^{(t+1)} = \theta^{(t)} + s[I(\theta^{(t)})]^{-1} \nabla \ell(\theta^{(t)}).$$

Example 7.1 (Logistic regression)

$y_i \sim \mathrm{Ber}(p_i)$ with $\operatorname{logit} p_i = \eta_i = x_i^\top \beta$. Then

$$\ell(\beta) = \sum_i \{y_i x_i^\top \beta - \log(1 + e^{x_i^\top \beta})\}, \qquad \nabla \ell = X^\top (y - p),$$

$$-\nabla^2 \ell = X^\top W X, \qquad W = \mathrm{diag}\{p_i(1 - p_i)\}.$$

The Hessian does not involve y, so $I(\beta) = -\nabla^2 \ell$: Newton and Fisher scoring coincide. The update rearranges into

$$\beta^{(t+1)} = (X^\top W^{(t)} X)^{-1} X^\top W^{(t)} z^{(t)}, \qquad z^{(t)} = X\beta^{(t)} + s(W^{(t)})^{-1}(y - p^{(t)}),$$

a weighted least squares fit to the **working responses** $z^{(t)}$: **iteratively reweighted least squares** (IRLS). Any weighted least squares solver therefore fits logistic regression.

Example 7.2 (Poisson regression)

$y_i \sim \mathrm{Poi}(\lambda_i)$ with $\log \lambda_i = x_i^\top \beta$: $\ell(\beta) = \sum_i \{y_i x_i^\top \beta - e^{x_i^\top \beta} - \log y_i!\}$, $\nabla \ell = X^\top (y - \lambda)$, $-\nabla^2 \ell = X^\top W X$ with $W = \mathrm{diag}(\lambda_i)$. Again the Hessian is free of y, and Newton = Fisher scoring = IRLS with $z = X\beta + W^{-1}(y - \lambda)$.

7.5 Generalized linear models

These coincidences are no accident. An **exponential dispersion family** has density

$$p(y \mid \eta, \phi) = \exp\Big\{\frac{y\eta - b(\eta)}{a(\phi)} + c(y, \phi)\Big\},$$

with natural parameter η, mean $\mu = b'(\eta)$ and variance $b''(\eta)a(\phi)$. A **GLM** links the mean to a linear predictor, $g(\mu) = x^\top\beta$; the **canonical link** is $g = (b')^{-1}$, which makes $\eta = x^\top\beta$.

family	canonical link	variance function
normal	$\eta = \mu$	1
Poisson	$\eta = \log\mu$	μ
binomial	$\eta = \log\frac{\mu}{1-\mu}$	$\mu(1-\mu)$
gamma	$\eta = \mu^{-1}$	μ^2
inverse Gaussian	$\eta = \mu^{-2}$	μ^3

Proposition 7.2

With $a(\phi) \equiv 1$, $\mu_i = g^{-1}(x_i^\top\beta)$ and $\sigma_i^2 = b''(\eta_i)$:

$$\nabla\ell(\beta) = \sum_i \frac{(y_i - \mu_i)}{\sigma_i^2 g'(\mu_i)} x_i, \qquad I(\beta) = \sum_i \frac{1}{\sigma_i^2 g'(\mu_i)^2} x_i x_i^\top = X^\top W X.$$

Fisher scoring is IRLS with weights $w_i = 1/\{\sigma_i^2 g'(\mu_i)^2\}$. For the canonical link, $-\nabla^2\ell = I(\beta)$, so Newton = Fisher scoring = IRLS and $-\ell$ is convex.

Proof. $\ell_i = y_i\eta_i - b(\eta_i)$, so $\partial\ell_i/\partial\eta_i = y_i - \mu_i$. By the chain rule, $d\mu = b''(\eta)d\eta = \sigma^2 d\eta$ and $d\mu = x^\top d\beta / g'(\mu)$, so $\partial\eta_i/\partial\beta = x_i/\{\sigma_i^2 g'(\mu_i)\}$, which gives the score. Differentiating again, one term is $-x_i x_i^\top/\{\sigma_i^2 g'(\mu_i)^2\}$ and the other is proportional to $(y_i - \mu_i)$, which has mean 0; this gives $I(\beta)$. For the canonical link $\eta_i = x_i^\top\beta$, so $\partial\eta_i/\partial\beta = x_i$ and $\nabla^2\ell = -\sum_i b''(\eta_i) x_i x_i^\top$, which involves no y_i and equals $-I(\beta)$ (here $\sigma_i^2 g'(\mu_i) = 1$). ■

For non-canonical links (e.g. probit) Fisher scoring and Newton differ and the likelihood need not be concave (probit happens to be log-concave). R's `glm` uses Fisher scoring.

Example 7.3 (Probit regression)

Let $y_i \sim \text{Ber}(p_i)$ with $p_i = \Phi(\eta_i)$, $\eta_i = \beta_0 + \beta_1 x_i$, and write $\Phi_i = \Phi(\eta_i)$, $\phi_i = \phi(\eta_i)$, $\mathbf{x}_i = (1, x_i)^\top$. Then

$$\ell(\beta) = \sum_i \{y_i \log\Phi_i + (1 - y_i)\log(1 - \Phi_i)\}, \qquad \nabla\ell = \sum_i \frac{(y_i - \Phi_i)\phi_i}{\Phi_i(1 - \Phi_i)} \mathbf{x}_i.$$

Differentiating ϕ_i/Φ_i and $-\phi_i/(1 - \Phi_i)$ in η with $\phi'(z) = -z\phi(z)$ gives

$$\nabla^2\ell = \sum_i f_i \mathbf{x}_i\mathbf{x}_i^\top, \qquad f_i = -y_i\frac{\phi_i(\eta_i\Phi_i + \phi_i)}{\Phi_i^2} - (1 - y_i)\frac{\phi_i\{\phi_i - \eta_i(1 - \Phi_i)\}}{(1 - \Phi_i)^2},$$

which depends on y_i: the link is not canonical. Taking expectations ($Ey_i = \Phi_i$) gives the Fisher information $X^\top W X$ with $w_i = \phi_i^2/\{\Phi_i(1 - \Phi_i)\}$, in agreement with Proposition 7.2 ($\sigma_i^2 = \Phi_i(1 - \Phi_i)$, $g'(\mu_i) = 1/\phi_i$). The IRLS step is

$$\beta^{(t+1)} = (X^\top W^{(t)} X)^{-1} X^\top W^{(t)} z^{(t)}, \qquad z_i^{(t)} = \eta_i^{(t)} + \frac{y_i - \Phi_i^{(t)}}{\phi_i^{(t)}}.$$

7.6 Nonlinear least squares: Gauss–Newton

Gauss located the dwarf planet Ceres in 1801 by least squares. In general, minimize

$$f(\beta) = \tfrac{1}{2}\sum_{i=1}^{n}\{y_i - \mu_i(\beta)\}^2,$$

e.g. a logistic growth curve $\mu(x;\beta) = \beta_3/(1+e^{-\beta_1-\beta_2 x})$. With $J(\beta)$ the $n \times p$ Jacobian whose rows are $\nabla\mu_i(\beta)^\top$,

$$\nabla f = -J^\top(y-\mu), \qquad \nabla^2 f = J^\top J - \sum_i (y_i - \mu_i)\nabla^2\mu_i.$$

Gauss–Newton drops the second term, using the positive semidefinite $J^\top J$:

$$\beta^{(t+1)} = \beta^{(t)} + s(J^\top J)^{-1}J^\top(y - \mu(\beta^{(t)})),$$

i.e. each step is a linear least squares fit of the residuals on J. It is justified when residuals are small or μ is nearly linear, and it is exactly Fisher scoring for the model $y_i \sim N(\mu_i(\beta), \sigma^2)$, since $E[-\nabla^2\ell] \propto J^\top J$. (Gauss introduced the normal distribution partly to justify least squares.)

7.7 Gradient descent

$$x^{(t+1)} = x^{(t)} - \gamma_t \nabla f(x^{(t)}).$$

Minimizing the linear model $\nabla f^\top \Delta x$ over $\|\Delta x\|_2 \le 1$ gives $\Delta x = -\nabla f/\|\nabla f\|$: the steepest descent direction. Because a linear model is unbounded below, a step size must always be chosen. Each iteration is cheap and needs no Hessian, which makes the method attractive at large scale; but it zigzags and converges slowly, and it does not apply to nonsmooth f.

Theorem 7.3

Let f be convex and differentiable with L-Lipschitz gradient, $\|\nabla f(x) - \nabla f(y)\|_2 \le L\|x-y\|_2$, and let f attain its minimum $p^\star$ at $x^\star$. With constant step $\gamma \in (0, 1/L]$,

$$f(x^{(t)}) - p^\star \le \frac{\|x^{(0)} - x^\star\|_2^2}{2\gamma t}.$$

Proof. The Lipschitz gradient gives the quadratic upper bound $f(y) \le f(x) + \nabla f(x)^\top(y-x) + \frac{L}{2}\|y-x\|^2$. With $y = x^+ = x - \gamma\nabla f(x)$ and $\gamma L \le 1$, $f(x^+) \le f(x) - \frac{\gamma}{2}\|\nabla f(x)\|^2$ (descent). By convexity, $f(x) \le p^\star + \nabla f(x)^\top(x - x^\star)$, so

$$f(x^+) - p^\star \le \nabla f(x)^\top(x-x^\star) - \frac{\gamma}{2}\|\nabla f(x)\|^2 = \frac{1}{2\gamma}(\|x-x^\star\|^2 - \|x^+ - x^\star\|^2).$$

Summing over t iterations telescopes: $\sum_{k=1}^{t}\{f(x^{(k)}) - p^\star\} \le \frac{1}{2\gamma}\|x^{(0)} - x^\star\|^2$. Since $f(x^{(k)})$ is nonincreasing, the left side is at least $t\{f(x^{(t)}) - p^\star\}$. ∎

This $O(1/t)$ rate is *sublinear*; with strong convexity one gets linear convergence, but never Newton's quadratic rate.

Example 7.4

- Least squares $f(\beta) = \frac{1}{2}\|X\beta - y\|^2$: $\nabla f = X^\top X\beta - X^\top y$ is Lipschitz with $L = \sigma_{\max}(X)^2$, so $\gamma \le 1/\sigma_{\max}(X)^2$ works.
- Logistic regression: $\nabla f = -X^\top(y-p)$ is Lipschitz with $L = \frac{1}{4}\|X^\top X\|_2$, since $p(1-p) \le$

$\frac{1}{4}$. The minimum may not exist (e.g. separable data); a ridge penalty $\frac{\rho}{2}\|\beta\|^2$ fixes this.

- Poisson regression: the gradient $-X^\top(y - e^{X\beta})$ is not globally Lipschitz, so a constant step size can fail; line search is needed. Indeed, an L-Lipschitz gradient forces $v^\top \nabla^2 f(\beta) v \le L\|v\|^2$ for all β, v. But for any v with $x_j^\top v > 0$ for some j,

$$v^\top \nabla^2 f(tv) v = \sum_i e^{t x_i^\top v} (x_i^\top v)^2 \ge e^{t x_j^\top v} (x_j^\top v)^2 \to \infty \quad (t \to \infty).$$

Example 7.5 (Multinomial logistic regression)

With K classes, dummy-coded responses $y_i \in \{0,1\}^K$, and $\log(p_{ik}/p_{iK}) = x_i^\top \beta_k$ ($k < K$, $\beta_K = 0$), we have $p_{ik} = e^{x_i^\top \beta_k}/(1 + \sum_{l<K} e^{x_i^\top \beta_l})$. With $B = [\beta_1, \ldots, \beta_{K-1}] \in \mathbb{R}^{p\times(K-1)}$, Y_{-K} the first $K-1$ columns of Y, and P the $n \times (K-1)$ matrix of the p_{ik},

$$\ell(B) = \sum_i \sum_k y_{ik} \log p_{ik} = \operatorname{tr}(Y_{-K}^\top X B) - \sum_i \log\Big(1 + \sum_{l<K} e^{x_i^\top \beta_l}\Big),$$

$$\nabla_{\beta_k} \ell = \sum_i x_i (y_{ik} - p_{ik}), \qquad \text{so} \qquad B^{(t+1)} = B^{(t)} + \gamma X^\top (Y_{-K} - P^{(t)})$$

is gradient ascent. For $K = 2$ it reduces to logistic regression. Each step costs $O(npK)$ and needs no $p(K-1)$-dimensional Hessian; the gradient is Lipschitz (the Hessian is bounded by $\frac{1}{2}\|X\|_2^2$ in operator norm), so a constant step works.

Example 7.6 (Quantiles by optimization)

The α-quantile of X (cdf F, positive density f) solves $F(x) = \alpha$. Newton's method is $x^{(t+1)} = x^{(t)} - \{F(x^{(t)}) - \alpha\}/f(x^{(t)})$. It is Newton's method for minimizing a function ψ with $\psi' = F - \alpha$ and $\psi'' = f > 0$. Since $F(x) = E[\mathbf{1}\{X \le x\}]$, integrating in x and interchanging with the expectation gives

$$\psi(x) = E\big[(x - X)\mathbf{1}\{X \le x\}\big] - \alpha x = E\big[\rho_\alpha(X - x)\big] + \text{const}, \qquad \rho_\alpha(u) = u\{\alpha - \mathbf{1}(u < 0)\},$$

the expected **check loss** of quantile regression; it is convex. Gradient descent reads $x^{(t+1)} = x^{(t)} - \gamma_t\{F(x^{(t)}) - \alpha\}$ and needs no density. If F is unavailable but X is easy to sample, replace the gradient by its unbiased one-sample estimate,

$$x^{(t+1)} = x^{(t)} - \gamma_t(\mathbf{1}\{X_t \le x^{(t)}\} - \alpha), \qquad X_t \overset{\text{iid}}{\sim} F,$$

with $\sum_t \gamma_t = \infty$, $\sum_t \gamma_t^2 < \infty$ (e.g. $\gamma_t = c/t$): this *stochastic gradient* (Robbins–Monro) algorithm converges to the quantile almost surely. The simplest Monte Carlo alternative is the sample α-quantile of N draws, which has standard error $\sqrt{\alpha(1-\alpha)/N}/f(x_\alpha)$.

7.8 Coordinate descent

Minimize over one coordinate (or block) at a time, cycling:

$$x_j^{(t+1)} = \arg\min_{x_j} f(x_1^{(t+1)}, \ldots, x_{j-1}^{(t+1)}, x_j, x_{j+1}^{(t)}, \ldots, x_d^{(t)}).$$

Each one-dimensional problem is often easy (sometimes closed form). The objective values decrease monotonically, so they converge. It is the optimization analogue of Gauss–Seidel. But even for

convex f, if f is not differentiable, coordinate descent can get stuck at a non-optimal point where no single coordinate move helps (a "corner" of a nonsmooth convex function).